

Environment

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1. a. What is environmental risk? Explain the risk management process regarding environmental degradation.
 b. How can a risk be leveled as acceptable or unacceptable by using risk matrix?
 c. Define residual risk. 3
2. a. Distinguish between risk and uncertainty. 3
 b. show how risk affects the environmental decision making. What is certainty equivalence? 10
 c. Explain the concept of option price and option value. 16
 d. Suppose you are thinking to buy a lottery and that lottery can give you \$100 with probability half or it can give you zero dollar with probability half right. Your utility function is 16

$$U = \sqrt{w}$$

- i. Are you risk averse?
- ii. What is certainty equivalence? Calculate risk premium.

3. a. Distinguish between flow and stock pollutants. Explain the efficient level of emission of stock pollutants. Explain the role of spatial differentiation for emission targets. 18
 b. Find the efficient solution for a uniformly mixed and short-lived stock pollutant with the following information: 7

$$B = 96M - 0.2M^2 \quad \text{for } 0 \leq M \leq 240$$

$$11520 \quad \text{for } M > 240$$

$$D = M^2 \quad \text{for } M \geq 0$$

4. a. Explain the criteria for evaluation of the pollution control instruments.
 B. Evaluate the arguments for the use of market or incentive-based instruments versus "command and control" instruments in the regulation environmental externalities under conditions of certainty.

Environment

Department of Economics
Jahangirnagar University
Tutorial Exam

Marks- 40

Time- 1.5 Hours

Answer any two the following Questions

1. a) What is Utilitarianism?

b) Critically explain the concept of preference-satisfaction utilitarianism as a basis for modern welfare economics.

2. a. In what sense Rawl's 'Theory of Justice' challenges classical utilitarianism? Explain.

b) Define externality. Suppose that a wood pulp mill is situated on a bank of a river. The marginal benefits and costs are as follows-

$$MC = 10 + 0.5Y$$

$$MB = 30 - 0.5Y$$

Where Y is tons of wood pulp produced. Each tons of wood pulp produces pollutant flows into the river which cause damage valued at \$10.

Find the socially efficient output level of wood pulp and the private profit maximizing output level. Why these two levels are different.

3. a) Define and explain Hedonic Pricing technique for environmental valuation. Why is it widely used?

b) What are the problems that arise in valuing the environment using the Hedonic Pricing technique?

4. a) Which type of environmental issues is addressed by travel cost method?

b) How does revealed preference become an important instrument to assess Travel cost method for environmental valuation?

c) Discuss why contingent valuation is superior to all other techniques in valuing environment.

5. a) Define compensating variation and equivalent variation in consumer preference.

b) Suppose that an individual has the utility function $U = E^{0.25} + Y^{0.75}$ where E is the some index of environmental quality and Y is income. From an initial situation where $E = 1$ and $Y = 100$, Calculate CS and ES for an increase in E to the level 2 and for a decrease in E to the level 0.5

$$\begin{array}{r} 40 - 10 \\ 1 - 10 \\ \hline 20 - 10 \\ \hline 10 \end{array}$$

2. How would you interpret variables statistically significant at the 5% level? 6

Environmental Economics

Department of Economics

Jahangirnagar University

M. Sc. Final Examination 2016

Course: Environmental Economics (Econ 506)

Time: 4 Hours Full Marks: 70

- Answer any five questions
- All parts of a question must be answered consecutively.

1. A. Environmental quality is a luxury good-explain views on that. (5)
B. Explain Environmental Kuznets Curve (EKC). (5)
C. Why do Dasgupta and his co-authors believe that the EKC can be lower and flatter? (4)
2. A. How would you identify the output level that is socially efficient? What is the difference between equity and efficiency? (6)
B. What are reasons for market failure when matters of environmental quality are involved. Imagine that a factory conveniently discharges its residuals into the nearby stream, causing harm to other users. Illustrate the probable outcomes under the private market system as against the socially efficient outcome. (8)
3. A. What is 'Utilitarianism'? (3)
B. How does Utilitarianism ensure maximization of social welfare for two individuals with different utility functions? (6)
C. How does Rawls's work challenge classical utilitarianism, explain. (5)
4. A. Derive the utilitarian intertemporal social welfare function. (4)
B. Why the future utility needs to be discounted? (3)
C. Optimal consumption growth path exhibits the interaction between impatience and productivity-derive its basic model. (4)
D. How do we find optimal growth with non-renewable resources used in production? (3)
5. A. Distinguish between 'sustainable' development, 'sustained' development and 'survivable' development which were explained by Pezzey (1997). (4)
B. To what extent are the ecologist's and the economist's concepts of sustainable behavior mutually consistent? (5)
C. Given that the question of the substitution possibilities as between human-made and natural capital is so important, how can the fact that we do not know the answer are explained? (5)
6. A. Define Hedonic Pricing technique for environmental valuation and explain briefly. Where is it widely used? (8)
B. What are the problems that arise in valuing the environment using this Hedonic Pricing technique? (6)
7. A. Which type of environmental issues is addressed by travel cost method? (4)
B. How does revealed preference become an important instrument to assess Travel cost (TC) method for environmental valuation? (10)
8. Discuss the contention that contingent valuation is, in general, superior to all other techniques for valuing non-marketed goods or services as it is the only technique capable of incorporating non-use values as well as use values. (14)

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Department of Economics

Jahangirnagar University

M. Sc. Final Examination 2017

Course Name: Environmental Economics (Econ 506)

Time: 4 Hours Full Marks: 70

- Answer any five questions
- All parts of a question must be answered consecutively.

1. A. Environmental quality is a luxury good- Explain. 4

B. Does economic growth inevitably lead to environmental degradation? How it can be explained with Environmental Kuznets Curve? How does this curve take different shapes for different countries? 10

2. A. How does ecological economics take a whole-systems approach to understand economic behavior as opposed to Environmental economics? 5

B. What is 'social rate of time preference' on a social indifference curve? 3

C. How do you derive the optimal combination of current and future consumption on a temporal production possibility curve? 6

3. A. Which type of environmental issues are addressed by Travel Cost Method (TCM)? 7

B. How does revealed preference become an important instrument to assess travel cost method for environmental valuation? 7

4. A. What is 'Anthropocentric utilitarianism'? 3

B. How does Utilitarianism explain that an unequal distribution of goods/consumption at a welfare maximum may occur? 5

C. How does Rawls's theory confront the welfare maximization concept of Utilitarianism? 6

5. A. Why should we work with positive time preference while discounting future utility? 5

B. Given the intertemporal consumption growth path, derive the optimal usage of both renewable and non-renewable resources. 9

6. A. Is sustainability feasible both for renewable and non-renewable resources? What are the arguments given by Robert Solow in this regard? 7

B. How does Hartwick rule become an important integral part in sustainability discussion? 7

7. A. Discuss the steps involved in applying Contingent Valuation Method. 9

B. Discuss the contention that, where use values are at issue, contingent valuation is superior to indirect methods as it goes directly to the appropriate theoretical construct for welfare analysis. 5

8. Short notes: (Any three)

A. Material Balance Method; B. Concepts of i) Sustainable development; ii) Sustained development and iii) Survivable development; C. Hedonic pricing; D. Hotelling rule; and E. Paris Agreement.

Department of Economics

Department of Economics
Jahangirnagar University
M. Sc. Final Examination 2018
Course: Environmental Economics (Econ 506)
Time: 4 Hours Full Marks: 70

*Department
of
Economics*

- Answer any five questions
- All parts of a question must be answered consecutively.

1. a) How do 'Environmental Kuznet's Curve' have a policy relevance in a growing economy to manage environmental degradation? 7
b) What is the social rate of time preference for consumption in the neo-classical growth model? Explain the concept. 7
2. a) How does preference-satisfaction utilitarianism differ from anthropocentric utilitarianism? 4
b) In positive economics? Economists have established preference ordering by ordinal utility functions, while the utilitarianism takes up cardinal utility function- explain. 4
c) How does an unequal distribution of resources among individuals at a social welfare maximum may occur? 6
3. a) Derive a Rawlsian social welfare function explaining the principles of justice. 5
b) What are the reasons for discounting future consumption? 3
c) Derive an optimal growth model with non-renewable resources used in production. 6
4. a) How do Jack Pezzy explain the concept of sustainability which can illuminate the difficulties of coming up with a single all-embracing definition? 7
b) 'The use of a non-renewable resource in production does not mean that sustainability as constant consumption is infeasible'- how do economists explain such perspective? 7
5. a) How does setting an environmental quality standard as legal limit ensures allocative efficiency to reduce pollution? 4
b) A national abatement standard cannot be optimal if there are regional difference – explain. 4
c) How does Pigouvian subsidy try to address the intangible benefits like improved health or more stable ecosystem? 6
6. a) Explain the risk assessment processes explaining an environment risk. 5
b) How do you define "acceptable risk" from an environment degradation? 4
c) What are the risk management strategies for a public policy to address acceptable risk? 5
7. a) Explain the guiding principles of the Paris Agreement. 4
b) Being a member in 'Least Development Country' (LDC) group, how do you assess Paris Agreement in terms of articles related to 'Adaptation'? 10
8. write short notes on : (any three) 14
 - a) Utilitarian intertemporal social welfare function
 - b) Weak and strong sustainability
 - c) Hartwick Rule
 - d) Environmental impact Assessment
 - e) Travel Cost Method
